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Solubility of carbon dioxide in eicosane, docosane, tetracosane, and octacosane at temperatures from 323 to 473 K and pressures up to 40 MPa

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Abstract

Solubilities of carbon dioxide in eicosane ($C_{20}H_{42}$), docosane ($C_{22}H_{46}$), tetracosane ($C_{24}H_{50}$), and octacosane ($C_{28}H_{58}$) were measured at temperatures from 323.2 to 473.2 K and pressures up to 40 MPa. The solubility increased with pressure and decreased with temperature. At a temperature of 323.2 K, bubble point pressures of carbon dioxide–eicosane and –docosane systems showed a large enhancement around 0.78 CO_2 mole fraction. At the higher temperatures, bubble point measurements were carried out up to the vicinity of the mixture critical points. Prediction of the solubility using the PSRK and the SRK equations of state with two excess Gibbs free energy mixing rules was examined. The mixing rule proposed by Zhong and Masuoka [C. Zhong, H. Masuoka, J. Chem. Eng. Jpn. 29 (1996) 315–322] provided prediction of the solubility with an average absolute deviation of 2.3% and can predict the solubility with satisfactory accuracy except near the mixture critical point. © 1998 Elsevier Science B.V. All rights reserved.

Keywords: Vapor–liquid equilibria; Equation of state; Mixing rule; Carbon dioxide; Alkane; High pressure

1. Introduction

Recently, various processes utilizing unique characteristics of supercritical fluids have been investigated. Information on phase equilibria for mixtures containing supercritical fluids is essential to realize these processes. These processes mainly treat mixtures containing highly asymmetric components in molecular size, such as the supercritical carbon dioxide extraction of heavy hydrocarbons.

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However, the accumulation of experimental data of these mixtures is not sufficient, and the development of accurate prediction methods is also desired.

Many prediction methods have been proposed for high pressure vapor–liquid equilibria [1–6]. The combination of an excess Gibbs free energy (G^E) model with an equation of state (EOS) was first proposed by Vidal [1], who equated the G^E from an EOS at infinite pressure with that from an existing G^E model. The method has been receiving much attention, and a number of attempts have been made to improve the predictive accuracy for complex systems [2–6]. Michelsen [5] proposed the modified Huron–Vidal (MHV1) mixing rule by deriving G^E from an EOS at zero pressure. This mixing rule can predict phase equilibria for mixtures of low molecular weight components accurately from the information on low pressures vapor liquid equilibria. Holderbaum and Gmehling [6] developed a predictive Soave–Redlich–Kwong (PSRK) EOS based on the SRK EOS [7] and a G^E mixing rule. It uses the UNIFAC method to calculate the mixture parameter and includes all already existing UNIFAC parameters. Additional PSRK parameters such as CO_2, which allow the calculation of gas/gas and gas/alkane phase equilibria, were determined [6,8]. However, Zhong and Masuoka [9] pointed out that the MHV1 mixing rule is not accurate for mixtures of gases (CO_2, CH_4, C_2H_6) and large alkanes with a carbon number higher than 10. Boukouvalas et al. [10] also reported the PSRK EOS showed a similar tendency, and developed the LCVM mixing rule to overcome this problem. Zhong and Masuoka introduced a correction factor, f, into the MHV1 mixing rule to improve the predictive accuracy for highly asymmetric gas–large alkane systems (MR1 mixing rule). The factor was almost a linear function of the carbon number of heavy alkanes depending only on lighter components. They tested the method for hydrocarbons up to carbon number 36 at pressures up to 10 MPa. A similar investigation was performed by Zhong et al. [11] for mixtures of gases (CO, H_2, C_2H_4) with large alkanes.

In this work, a new apparatus was developed based on a synthetic method. The maximum allowable operating temperature and pressure are 473 K and 100 MPa, respectively. Solubility of carbon dioxide in eicosane ($C_{20}H_{42}$), docosane ($C_{22}H_{46}$), tetracosane ($C_{24}H_{50}$), and octacosane ($C_{28}H_{58}$) were measured at temperatures from 323.2 K to 473.2 K and pressures up to 40 MPa by using the apparatus. The experimental results were compared with predictions by the PSRK EOS and SRK EOS with two mixing rules, i.e., MHV1 mixing rule and MR1 mixing rule.

2. Theory

The SRK EOS [7] was used for the prediction of vapor–liquid equilibria in this work:

$$P = \frac{RT}{v-b} - \frac{a}{v(v+b)} \tag{1}$$

For pure components, constants a and b are evaluated by

$$a = 0.42748\frac{R^2T_c^2}{P_c}\beta \tag{2}$$

$$b = 0.08664\frac{RT_c}{P_c} \tag{3}$$

$$\beta = \left[1 + m\left(1 - T_R^{0.5}\right)\right]^2 \tag{4}$$

and

$$m = 0.48508 + 1.55171\omega - 0.15613\omega^2 \tag{5}$$

where Eq. (5) was developed by Graboski and Daubert [12] using vapor pressure data of methane to eicosane and was used for extrapolation to heavier paraffins in this work. In case of the PSRK, the following expression proposed by Mathias and Copeman [13] was used:

$$\beta = \left[1 + c_1\left(1 - T_R^{0.5}\right) + c_2\left(1 - T_R^{0.5}\right)^2 + c_3\left(1 - T_R^{0.5}\right)^3\right]^2 \tag{6}$$

Parameters c_1, c_2, and c_3 of alkanes were determined by fitting vapor pressure data [14] and those of CO_2 were obtained from Holderbaum and Gmehling [6]. These parameters are tabulated in Table 1.

Values of T_c, P_c, and ω for the alkanes except for eicosane were predicted by the method of Gasem and Robinson's correlation as described by Huang et al. [15]. Those of carbon dioxide and eicosane were taken from the property data book of Reid et al. [16]. Those values are also tabulated in Table 1.

Appropriate mixing rules are needed to calculate thermodynamic properties for mixtures. Michelsen [5] derived a G^E mixing rule (MHV1) by equating the G^E from an EOS at zero pressure to that from an activity coefficient model. Zhong and Masuoka [9] proposed a modified MHV1 mixing rule (MR1) for carbon dioxide–alkane system by introducing a correction factor, f, as follows:

$$\alpha_m = \sum_i x_i \alpha_i + (1/q_1)\left[(G^E/RT)(1+f) + \sum_i x_i \ln(b_m/b_i)\right] \tag{7}$$

$$f = 0.215 + 0.00375n \qquad (n > 10) \tag{8}$$

$$f = 0 \qquad (n \le 10) \tag{9}$$

$$b_m = \sum_i x_i b_i \tag{10}$$

where x_i is mole fraction of component i, $\alpha_m = a_m/(b_m RT)$, $\alpha_i = a_i/(b_i RT)$, and n is the carbon number of alkanes. Eqs. (8) and (9) are used in the MR1 mixing rule, while f is always set to be zero in the MHV1 mixing rule. In Eq. (7), q_1 is an EOS-dependent parameter that has the value of -0.64663 for SRK EOS as obtained by Holderbaum and Gmehling [6].

Table 1
Pure compound parameters [6,15,16]

Substance		T_c (K)	P_c (MPa)	ω (−)	c_1^a	c_2^a	c_3^a
Carbon dioxide	CO_2	304.1	7.38	0.239	0.8252	0.2515	−1.7039
Eicosane	$C_{20}H_{42}$	767.0	1.11	0.907	1.8410	−0.7811	1.3911
Docosane	$C_{22}H_{46}$	790.4	1.01	0.927	1.8216	−0.2040	0.4487
Tetracosane	$C_{24}H_{50}$	809.5	0.93	0.987	1.9228	−0.4866	1.2946
Octacosane	$C_{28}H_{58}$	843.5	0.82	1.096	2.0164	0.0120	−0.0009

[a] Determined in this work except for CO_2.

The UNIFAC model [17] was used in this work to predict the G^E. The volume and area parameters for CO_2 and the temperature-dependent interaction parameters for the CO_2–CH_2 group pair in the UNIFAC model were obtained from Holderbaum and Gmehling [6]. The temperature-dependent interaction parameter is expressed as:

$$\psi_{nm} = \exp\left(- \frac{a_{nm} + b_{nm}T + c_{nm}T^2}{T} \right) \tag{11}$$

3. Experimental

Eicosane ($C_{20}H_{42}$, purity $> 99.0\%$) and docosane ($C_{22}H_{46}$, purity $> 98.0\%$) were obtained from Tokyo Kasei. Tetracosane ($C_{24}H_{50}$, purity $> 99.0\%$) and octacosane ($C_{28}H_{58}$, purity $> 98.0\%$) were obtained from Katayama Chemicals. Carbon dioxide (purity $> 99.5\%$) was supplied from Iwatani Industrial Gases. All compounds were used without further purification.

The so-called synthetic method was used for the measurement of the solubility of CO_2 in heavy alkanes, i.e., bubble point pressures were determined at various temperatures for mixtures of known compositions. A schematic diagram of the apparatus used in this work is shown in Fig. 1. It consists of a high-pressure variable-volume view cell, a temperature controller, a hand pump, a pressure gauge, a position sensing device for the piston, and an optical observation system.

The cell (Hikari Koatsu) consisted of two sapphire windows (ϕ15, 10 mm thick) and a valve, and was constructed of 316 stainless steel. The maximum allowable operating condition was 473 K and 100 MPa. The inner volume of the cell could be varied from 8 to 15 cm^3 by moving the piston. The

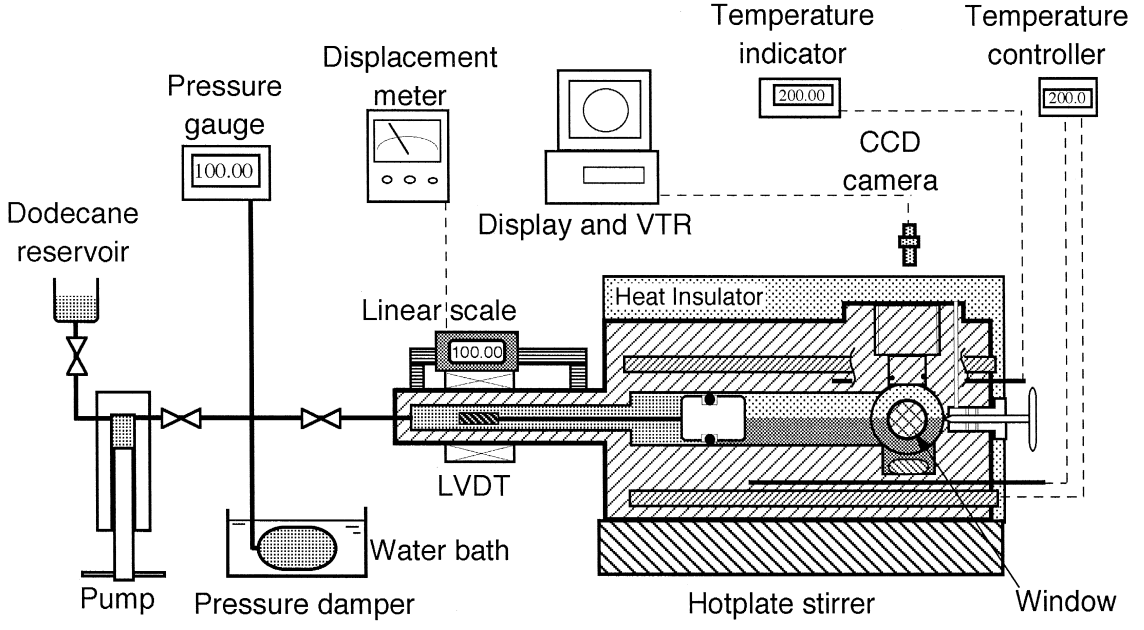


Fig. 1. Schematic diagram of the apparatus.

piston sensor consisted of a linear variable differential transformer (LVDT, Shinko Electric, model 6020) and a digital linear scale (Mitsutoyo, model SD-10B). A nonmagnetic 316 stainless steel rod connected the rear of the piston and a core of an LVDT. The stainless steel rod and the LVDT core were all housed in a nonmagnetic 316 stainless steel tube. The position of the piston was determined by the linear scale as well as the LVDT as a null detector with an accuracy of ± 0.03 mm. A Teflon-coated stirring bar was activated to provide mixing by a hot-plate stirrer which was located under the cell.

The cell was heated using four 200 W cartridge heaters. The heaters were wired to an AC power transformer and a PID controller (Shimaden, model SR22) which used a Pt 100 Ω RTD probe as an input sensor. This system allowed the inner bore temperature to be controlled to within an accuracy of ± 0.05 K. The temperature was measured to within ± 0.05 K with a three-wire class A RTD probe (Pt 100 Ω, Okazaki R34) and an indicator (Yokogawa Electric, model 7563). The RTD as well as the indicator were calibrated against an intelligent standard RTD probe (Kaye Instruments, model X0860, accuracy ± 0.01 K).

The sample in the cell was pressurized by a pressure-transmitting fluid (dodecane) using a hand pump (Hikari Koatsu). The pressure of dodecane was read on a precision Heise gauge (Dresser Industries, model 901A, F.S. 100 MPa), which had been calibrated against a dead weight tester (Pressurements, model M2800). The accuracy of the Heise gauge was within ± 0.01 MPa below 40 MPa. The accuracy of pressure in the cell is estimated within ± 0.08 MPa which includes the error due to pressure difference across the piston caused by the friction of a Teflon o-ring.

A sample of known composition was prepared gravimetrically as follows. The empty cell was purged several times with CO_2 to remove traces of air. Then, solid alkane was loaded into the cell through a top side bore with an accuracy within ± 0.4 mg. At the loading of alkane, a small amount of CO_2 was leaked through the bore to prevent air contamination. Then carbon dioxide was introduced into the cell through a filling tube from the sample cylinder. The carbon dioxide remaining in the filling tube was recovered in the sample cylinder with condensation using dry ice and methanol.

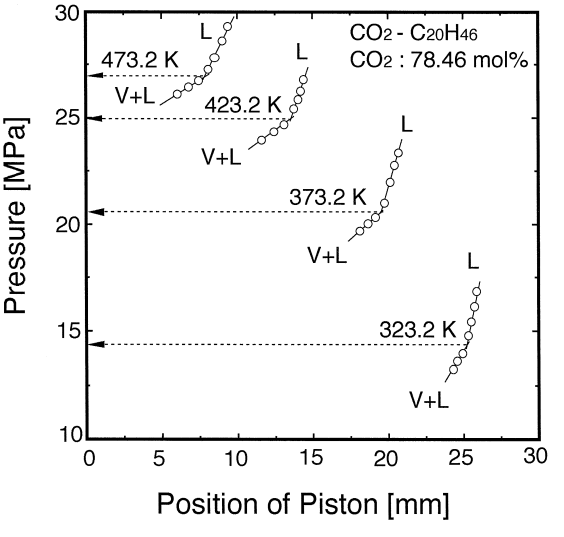


Fig. 2. Relationship between pressure and position of piston.

The amount of carbon dioxide introduced was determined by measuring the difference in masses of the sample cylinder before and after the introduction and by considering also pre-existing amount (about 10–20 mg) in the cell. Two types of the sample cylinders were used. One was about 350 g in mass and about 10 cm^3 in inner volume, and the other is 190 g and 2.5 cm^3. The latter was for samples below 0.6 CO_2 mole fraction. The mass of carbon dioxide that filled the cell varied from 2 to 5 g and was measured to within ± 4 mg and ± 1.0 mg, above and below 0.6 mole fraction, respectively. The uncertainty of the sample composition is estimated to be within ± 0.05 mol%.

Bubble point pressures were determined from the relationship between the pressure and the inner volume of the cell. In the operation, a sample of a known composition loaded in the cell was pressurized into a one-phase homogeneous solution by advancing the piston. After equilibration, the pressure was reduced stepwise at a constant temperature. The pressure and the position of the piston were recorded after each step as soon as the equilibrium had been reached. Fig. 2 shows a typical

Table 2
Solubilities of carbon dioxide in eicosane, docosane, tetracosane, and octacosane

Substance	x_{CO_2}	P (MPa)			
		323.2 K	373.2 K	423.2 K	473.2 K
Eicosane	0.5100	5.91	8.79	10.92	12.21
	0.6054	7.62	11.66	14.55	16.21
	0.7011	9.90	15.75	19.44	21.45
	0.7846	14.78	20.63	25.00	27.14
	0.8602		26.36	29.85	32.01
	0.9202		30.78	33.57	34.41
Docosane	0.5708	6.37	9.61	12.17	13.58
	0.6977	9.14	15.07	19.00	21.35
	0.7153	10.42	16.29	20.21	22.17
	0.7679	12.38	18.98	23.12	25.51
	0.8003		22.22	26.58	28.63
	0.8498		25.72	29.65	31.68
	0.8710		31.05	33.30	35.18
	0.8821		31.49	33.93	35.50
	0.9238		35.21	36.64	37.30
Tetracosane	0.5587		9.43	12.28	14.09
	0.5609		9.85	12.31	13.86
	0.6588		13.77	17.22	19.50
	0.6727		14.39	18.10	19.54
	0.7003		15.19	18.98	20.77
	0.8310		23.92	28.94	31.13
	0.8997		38.94	38.48	
Octacosane	0.5477		8.97	11.50	
	0.7298		16.19	21.82	
	0.7548		18.14	22.94	
	0.7990		22.28	27.09	
	0.8972			40.85	

result of a series of isothermal expansions. The measured pressures were plotted against the position of the piston. At the bubble point pressure, a change in the slope of the pressure with the position was clearly observed. The pressures measured in the vapor–liquid coexisting phase and the liquid phase were separately correlated by a linear approximation. The bubble point pressure was determined as an intersection of the two linear lines. The average reproducibility of the bubble point pressure was ± 0.06 MPa.

4. Results and discussion

The solubilities of carbon dioxide in eicosane and docosane were measured at pressures up to 40 MPa and four temperatures of 323.2, 373.2, 423.2, and 473.2 K. The solubilities of carbon dioxide in tetracosane and octacosane were measured at temperatures of 373.2, 423.2, and 473.2 K. The solubilities are listed in Table 2 and shown in Figs. 3–6. For the carbon dioxide–tetracosane and –octacosane systems, measurements were only carried out above the hydrocarbon melting point of 327 K and 337.7 K [18] for tetracosane and octacosane, respectively. At 473 K and 0.8997 CO_2 mole fraction for the CO_2–tetracosane system, the bubble point pressure could not be determined, because the amount of the sample in the cell was too large to form a vapor–liquid coexisting phase. In all systems, solubility data obtained in this work showed a reasonable connection with previous results [19–22].

In all systems, the solubility increased with decreasing temperature and the slopes of solubility against pressure slightly increased with CO_2 mole fraction. At 323.2 K, the bubble point pressures of the eicosane and docosane systems showed a large enhancement around 0.78 mole fraction. Formation of a one-phase homogeneous solution could not be observed up to 100 MPa at CO_2 mole

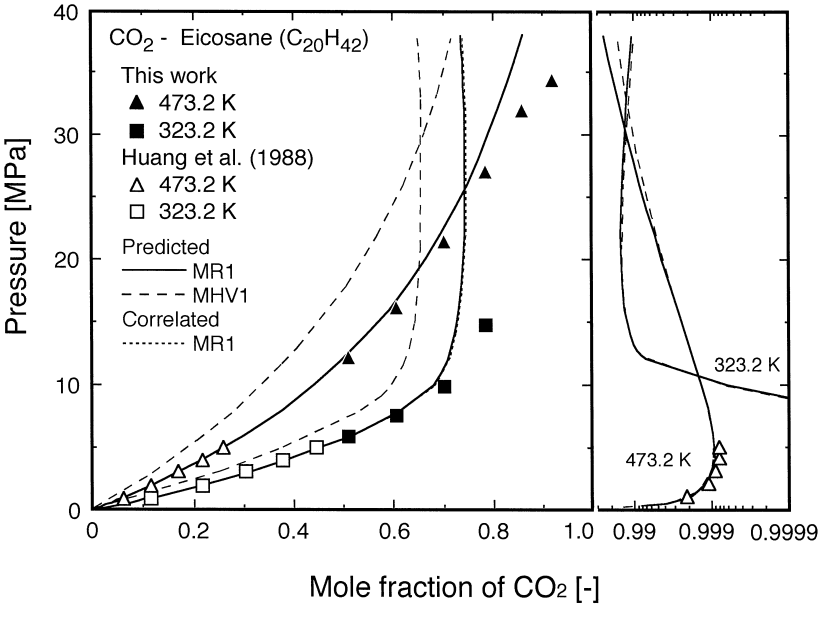


Fig. 3. Prediction and correlation result for the carbon dioxide–eicosane system.

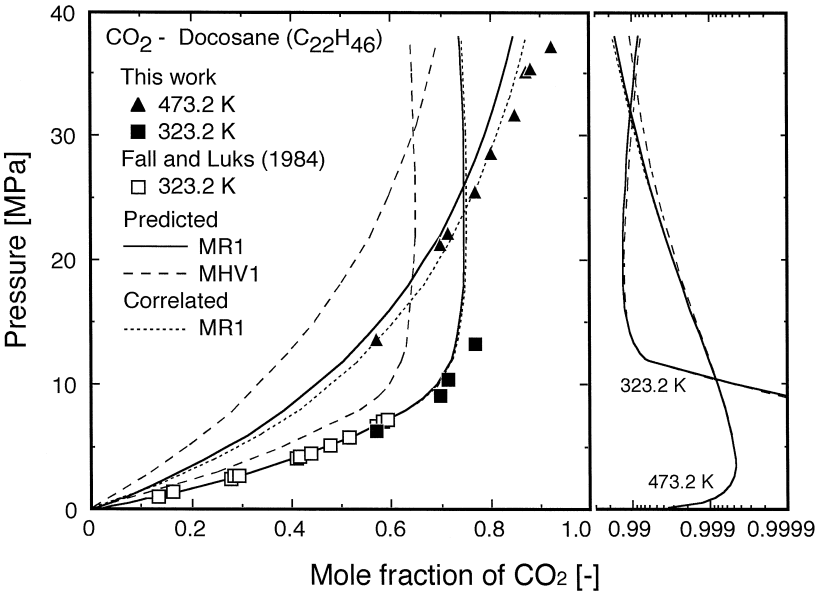


Fig. 4. Prediction and correlation result for the carbon dioxide–docosane system.

fractions higher than 0.8. Therefore, critical pressures of these mixtures at 323.2 K will be higher than 100 MPa. In the octacosane system, a one-phase homogeneous solution could not be formed up to 100 MPa at a temperature of 373.2 K at 0.85 mole fraction CO_2. This result is consistent with critical loci reported by Liphard and Schneider [23]. On the other hand, critical opalescence was observed

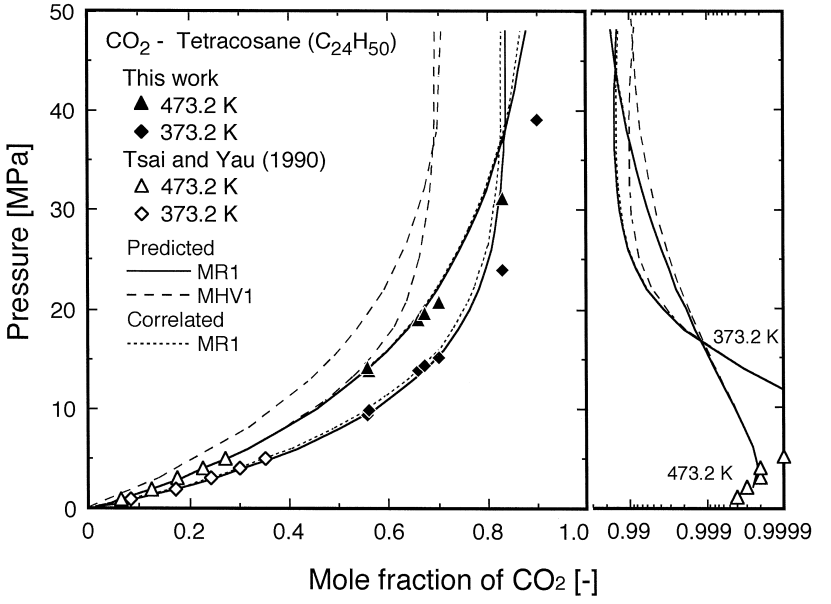


Fig. 5. Prediction and correlation result for the carbon dioxide–tetracosane system.

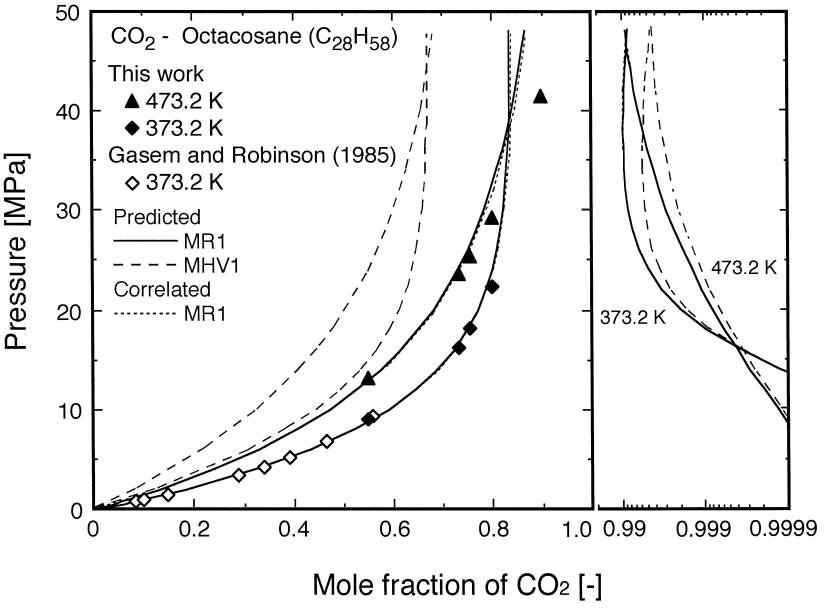


Fig. 6. Prediction and correlation result for the carbon dioxide–octacosane system.

around 0.92 mole fraction for the carbon dioxide–eicosane and –docosane systems at temperatures of 373.2 to 473.2 K. The bubble point pressures around 0.92 mole fraction would be near the critical point pressures of the mixtures. A comparison of bubble point pressures of the present systems at 423.2 K is shown in Fig. 7. While the bubble point pressures of these mixtures showed almost the same value at lower pressures, some differences can be seen at CO_2 mole fractions greater than 0.8. Similar results were obtained at other temperatures.

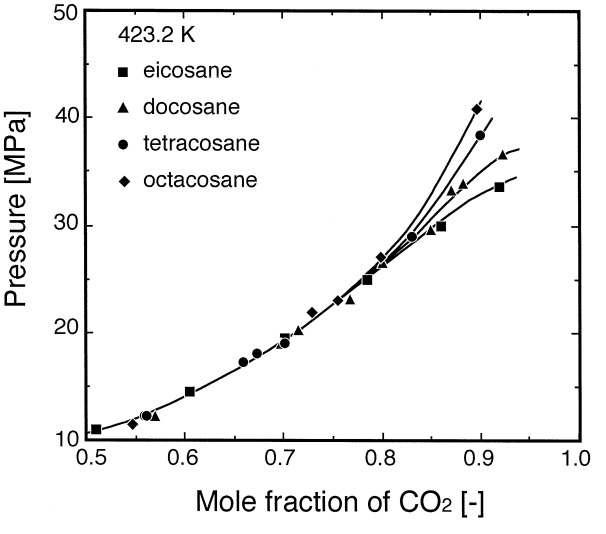


Fig. 7. Comparison of bubble point pressures for CO_2–alkane mixtures at 423.2 K.

Experimental results were compared with predictions using SRK equation of state. Average absolute deviations (AAD) by using three prediction methods, i.e., SRK EOS with MHV1 and MR1 mixing rules and PSRK EOS, are shown in Table 3. The deviations of PSRK EOS were as large as those of MHV1 mixing rule and became large with an increase in the alkane carbon number. An overall AAD of MR1 mixing rule was 2.25%.

Figs. 3–6 show the comparisons for the four systems, including predictions in vapor phase. Since results of PSRK EOS were similar to those of MHV1 mixing rule, two mixing rules were compared in the prediction of bubble point pressures in Figs. 3–6. Broken lines denote predicted values using MHV1 mixing rule, and solid lines represent those by MR1 mixing rule. MHV1 mixing rule predicted higher bubble point pressures, i.e., lower solubilities of supercritical CO_2 in the alkanes at high pressure than the experimental values for all systems examined. Especially, the deviations of predicted values using MHV1 from experimental values became large with an increase in the alkane carbon number. On the other hand, MR1 mixing rule showed quantitative agreement within the experimental data except in the vicinity of the mixture critical points.

Correlation of the measured solubility data was investigated by treating f in the MR1 mixing rule as a fitting parameter. Optimal f values were determined by the minimization of the difference between measured and calculated CO_2 mole fractions in the liquid phase at each temperature. Experimental data in the literatures [19–22] are included in the correlation. The correlated results are

Table 3
Prediction results by PSRK EOS and SRK EOS with MHV1 and MR1 mixing rules and correlation results with MR1 mixing rule for solubilities of carbon dioxide in alkanes

Alkane	Prediction				T (K)	Correlation (MR1)		
	MHV1	PSRK	MR1			f (−)	AAD[a] (%)	
	AAD[a] (%)	AAD[a] (%)	AAD[a] (%)	f (−)			Liquid	Vapor
Eicosane	20.5	18.7	2.30	0.2900	323.2	0.3049	1.94	
					373.2	0.2816	2.42	
					423.2	0.3334	2.43	
					473.2	0.2936	2.71	0.03
Docosane	21.0	21.3	2.40	0.2975	323.2	0.3153	1.40	
					373.2	0.3144	2.06	
					423.2	0.3583	1.90	
					473.2	0.3866	2.22	
Tetracosane	24.0	24.5	2.07	0.3050	373.2	0.2782	2.82	
					423.2	0.3086	1.89	
					473.2	0.2997	1.55	0.01
Octacosane	28.9	28.6	2.24	0.3200	373.2	0.3289	2.03	
					423.2	0.3363	2.09	
					473.2	0.3377	2.16	
Overall	23.6	23.3	2.33				2.25	0.02

[a]$\mathrm{AAD} = 100\ \Sigma(|z_{cal} - z_{exp}|/z_{exp})/n$.
z: mole fraction of CO_2.
n: number of data.

shown by dotted lines in Figs. 3–6. The correlated solubilities almost coincide with the predictions by MR1 mixing rule, especially in the vapor phase the correlated results show little difference from prediction. Table 3 lists AAD for the predictions and the correlations as well as f values. This result indicates that Eq. (8) gives values very close to the optimal ones for each alkane and that the prediction using the MR1 mixing rule is satisfactorily accurate for vapor–liquid equilibria examined in the present systems up to high pressures.

5. Conclusion

Solubilities of carbon dioxide in eicosane, docosane, tetracosane, and octacosane were measured at pressures up to 40 MPa and temperatures from 323.2 to 473.2 K. In all systems, the solubility increased with decreasing temperature and the slopes of the pressure slightly against the solubility increased with the mole fraction of CO_2. The solubilities of present systems showed almost the same value at a given temperature except for CO_2 mole fractions greater than 0.8.

Prediction of the solubility was carried out using PSRK and SRK equations of state with MHV1 mixing rule and MR1 mixing rule. While the PSRK and the SRK with the MHV1 mixing rule predicted higher bubble point pressures than the experimental values, prediction using the MR1 mixing rule was in good agreement with the experimental data, except near the critical point. Correlation of the measured solubility data was further investigated by treating 'f' in the MR1 mixing rule as a fitting parameter. The correlation results, however, show very little difference with prediction ones and that the MR1 mixing rule can predict the vapor–liquid equilibria with satisfactory accuracy for CO_2–heavy alkane systems up to 40 MPa.

6. List of symbols

a	parameter in SRK EOS	Pa m^6/mol^2
a_{nm}	group interaction parameter in UNIFAC equation	K
b	parameter in SRK EOS	m^3/mol
b_{nm}	group interaction parameter in UNIFAC equation	–
c_{nm}	group interaction parameter in UNIFAC equation	K^{-1}
c_1, c_2, c_3	parameter in Eq. (6)	–
f	correction parameter defined in Eq. (7)	–
G^E	excess Gibbs free energy	J/mol
m	parameter defined in Eq. (5)	–
n	carbon number of alkane	–
P	pressure	Pa
q_1	parameter in Eq. (7)	–
R	gas constant (8.314)	J/(mol K)
T	temperature	K
T_R	reduced temperature	–
v	molar volume	m^3/mol
x	mole fraction	–

Greek letters		
α	parameter in MHV1 and MR1	–
β	Parameter defined in Eqs. (4) and (6)	–
Ψ_{nm}	interaction energy parameter in UNIFAC equation	–
ω	acentric factor	–
Subscripts		
c	critical property	
cal.	calculated value	
exp.	experimental value	
i	component *i*	
m	mixture	
R	reduced property	

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References

[1] J. Vidal, Chem. Eng. Sci. 33 (1978) 787–791.
[2] M.J. Huron, J. Vidal, Fluid Phase Equilibria 3 (1979) 255–271.
[3] J. Mollerup, Fluid Phase Equilibria 25 (1986) 323–326.
[4] M.L. Michelsen, Fluid Phase Equilibria 60 (1990) 47–58.
[5] M.L. Michelsen, Fluid Phase Equilibria 60 (1990) 213–219.
[6] T. Holderbaum, J. Gmehling, Fluid Phase Equilibria 70 (1991) 251–265.
[7] G. Soave, Chem. Eng. Sci. 27 (1972) 1197–1203.
[8] K. Fischer, J. Gmehling, Fluid Phase Equilibria 121 (1996) 185–206.
[9] C. Zhong, H. Masuoka, J. Chem. Eng. Jpn. 29 (1996) 315–322.
[10] C. Boukouvalas, N. Spiliotis, P. Coutsikos, N. Tzouvaras, D. Tassios, Fluid Phase Equilibria 92 (1994) 75–106.
[11] C. Zhong, M. Takeuchi, H. Masuoka, J. Chem. Eng. Jpn. 30 (1997) 1133–1137.
[12] M.S. Graboski, T.E. Daubert, Ind. Eng. Chem. Process Des. Dev. 17 (1978) 443–448.
[13] P.M. Mathias, T.W. Copeman, Fluid Phase Equilibria 13 (1983) 91–108.
[14] D.L. Morgan, R. Kobayashi, Fluid Phase Equilibria 97 (1994) 211–242.
[15] S.H. Huang, H.M. Lin, F.N. Tsai, K.C. Chao, Ind. Eng. Chem. Res. 27 (1988) 162–169.
[16] R.C. Reid, J.M. Prausnitz, B.E. Poling, The Properties of Gases and Liquids, 4th edn., McGraw-Hill, New York, 1987.

[17] A. Fredenslund, R.L. Jones, J.M. Prausnitz, AIChE J. 21 (1975) 1086–1099.
[18] R.C. Weast (Ed.), Handbook of Chemistry and Physics, CRC Press, Boca Raton, FL, 1988, pp. C-380 and C-507.
[19] S.H. Huang, H.M. Lin, K.C. Chao, J. Chem. Eng. Data 33 (1988) 145–147.
[20] D.J. Fall, K.D. Luks, J. Chem. Eng. Data 29 (1984) 413–417.
[21] F.N. Tsai, J.S. Yau, J. Chem. Eng. Data 35 (1990) 43–45.
[22] K.A.M. Gasem, R.L. Robinson Jr., J. Chem. Eng. Data 30 (1985) 53–56.
[23] K.G. Liphard, G.M. Schneider, J. Chem. Thermodyn. 7 (1975) 805–814.